MONITORING & EVALUATION • HEALTH SYSTEMS PERFORMANCE PORTFOLIO

Health Facility Performance, Service Readiness, and Equity in Maternal and Child Health Service Delivery in Ethiopia

An M&E Analysis of the Ethiopia Service Provision Assessment (SPA) 2021–22

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Independent M&E / Health-Systems Data Analysis Portfolio Project

Analysis conducted in R • tidyverse, haven, labelled, ggplot2 • survey-weighted descriptive analysis

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1. Executive Summary

Health facility data are often reduced to a single question: does a facility offer a service? This project shows why that question is incomplete. Using the Ethiopia Service Provision Assessment (SPA) 2021–22 — a nationally representative, weighted facility survey — this analysis examines antenatal care (ANC), normal delivery, and child health services across three linked dimensions: whether services are available, whether facilities are actually ready to deliver them, and whether that readiness is distributed equitably across regions, residence, facility type, and managing authority.

The analytic sample includes 1,158 facilities with positive survey weights, drawn from 1,407 originally selected. All estimates are survey-weighted to reflect Ethiopia's stratified facility sampling design.

Three findings anchor this report:

- ANC is widely available (74.7%) but readiness lags far behind (61.2%), and readiness varies enormously by region — from 94.9% in Addis Ababa to 47.1% in Gambella.
- Normal delivery is less commonly available (53.5%, among facilities eligible to provide it) but shows the strongest readiness of the three domains (87.1%) once a facility offers the service.
- Child health services are nearly universal (89.7% availability) but readiness (66.6%) and facility-type disparities remain substantial.

Across all three domains, a cross-cutting infrastructure gap stands out: running water and soap — the most basic infection-prevention inputs — are the weakest readiness components almost everywhere they are assessed.

> **Main M&E implication:** Counting whether a service exists is not sufficient. Monitoring systems need to track availability and readiness together, and disaggregate both by region and facility type, to detect where investment is genuinely needed.

2. Background and Problem Statement

Health system performance is commonly tracked through service availability — the share of facilities that offer a given service. This is a necessary but incomplete indicator. A facility can report offering antenatal care, normal delivery, or child health services without holding the basic supplies, equipment, or infrastructure needed to deliver that service safely and consistently.

Service readiness captures this second dimension: whether a facility that offers a service actually has the tracer items required to provide it. Tracking readiness alongside availability reveals a more honest picture of what a health system can deliver, rather than what it claims to deliver.

Equity is the third dimension this project examines. National averages can mask sharp differences between regions, between urban and rural facilities, between facility types, and between public and non-government providers. A national readiness figure that looks acceptable can still conceal pockets of severe under-preparedness.

ANC, normal delivery, and child health were selected because they represent the core continuum of maternal and child health care — pregnancy, birth, and the postnatal/early-childhood period — and because health-system weaknesses at any of these three points carry direct consequences for maternal and child survival.

3. M&E Questions

This analysis is organized around five monitoring and evaluation questions:

1. How widely are ANC, normal delivery, and child-health services available across Ethiopian health facilities?

2. How ready are facilities to provide these services?

3. How does readiness vary across regions, residence, facility groups, and managing authority?

4. Which facility-level components represent the largest readiness bottlenecks?

5. What are the implications for health-system monitoring and quality improvement?

4. Data Source and Study Design

This analysis uses the Ethiopia Service Provision Assessment (SPA) 2021–22, a facility-based cross-sectional survey designed to assess health-facility infrastructure, service availability, service readiness, and quality-related aspects of care delivery.

Sample

- 1,407 facilities were selected for the SPA.
- 1,158 facilities had successfully collected, usable facility data and positive analytical survey weights; this is the analytic sample used throughout this report.
- 249 selected facilities had zero or missing analytical weight and lacked the core facility/service variables required for analysis; they are excluded from all weighted estimates.

Sampling Design and Survey Weights

The SPA uses a stratified sampling design, with strata defined by region and facility type (with additional stratification among clinics). Because facilities were not sampled proportionally across strata, all estimates in this report are survey-weighted using the facility sample weight supplied with the SPA (variable v005).

For this analysis, an analysis weight was created as analysis_weight = v005 ÷ 1,000,000. This normalization does not change weighted means or proportions; it simply rescales the weights so that the positive weights sum to 1,158 (i.e., a mean weight of 1). This is a normalized relative survey weight, not a population expansion factor.

Analytic Sample by Domain

Domain	Facilities Reporting the Service
ANC	905
Normal Delivery	650
Child Health	1,078

These counts represent facilities that reported offering the relevant service. As explained in Section 5, the denominator used to calculate service availability is not always identical to this count — this distinction is most important for normal delivery.

5. Variables and Indicator Construction

5.1 Service Availability

Service availability is the weighted proportion of eligible facilities offering a specified service.

- ANC: weighted proportion of analytic facilities offering ANC.
- Normal delivery: weighted proportion of eligible facilities — excluding health posts — offering normal delivery. Health posts are excluded from this denominator because they are not part of the official SPA normal-delivery availability indicator.
- Child health: weighted proportion of analytic facilities offering child-health services.

> **Denominator note:** An initial all-facility calculation for normal delivery produced 18.7%. This is not the final indicator. Once health posts are excluded from the denominator — consistent with the official SPA approach — weighted normal-delivery availability is 53.5%, which closely reproduces the official national SPA estimate of approximately 54%. This report uses 53.5% throughout.

5.2 Service Readiness

Readiness was constructed from selected, observable facility-level tracer components for each domain. The readiness score for a facility is the percentage of selected components available, summarized across facilities using the survey-weighted mean.

> **Important:** These are constructed service-readiness indices built for this portfolio analysis from a defined set of tracer components. They are not the official SPA readiness indices and should not be interpreted as such.

5.3 Readiness Components by Domain

ANC (2 components)	Normal Delivery (9 components)	Child Health (9 components)
Privacy	Privacy	Privacy
Running water	Running water	Running water
	Soap	Soap
	Gloves	Infant scale
	Suture	Scale
	Scissors	Thermometer
	Delivery pack	Stethoscope
	Cord clamp	IMNCI guideline
	Delivery bed	IMNCI chart

The ANC readiness index is built from only two components (privacy and running water), which makes it a comparatively simple readiness measure relative to the nine-component indices used for delivery and child health; this is a limitation the report returns to in Section 13.

5.4 Child-Health Readiness Completeness Rule

Child-health readiness was calculated only for facilities with at least 75% of the nine selected components assessed (i.e., at least 7 of 9). One facility with insufficient component information was excluded from the child-readiness calculation.

5.5 Equity Variables

Equity in readiness was assessed across four dimensions: region, urban/rural residence, facility group (Hospital, Health Center, Health Post, Clinic), and managing authority (public vs. non-government). Equity gaps are calculated as the difference between the highest and lowest weighted readiness estimate within each dimension.

6. Analytical Approach

- Service availability was estimated as a survey-weighted proportion of eligible facilities offering each service.
- Service readiness was estimated as a survey-weighted mean of the facility-level readiness score for each domain.
- Equity was assessed through stratified comparisons of weighted readiness across region, residence, facility group, and managing authority; equity gaps are the maximum minus minimum weighted readiness within each stratification.
- Component-level bottleneck analysis examined the weighted availability of each individual readiness component to identify which specific inputs are most frequently missing.
- All analyses were descriptive and survey-weighted; no regression modeling, significance testing, or causal analysis was performed.

The analytical workflow was developed in R and RStudio.

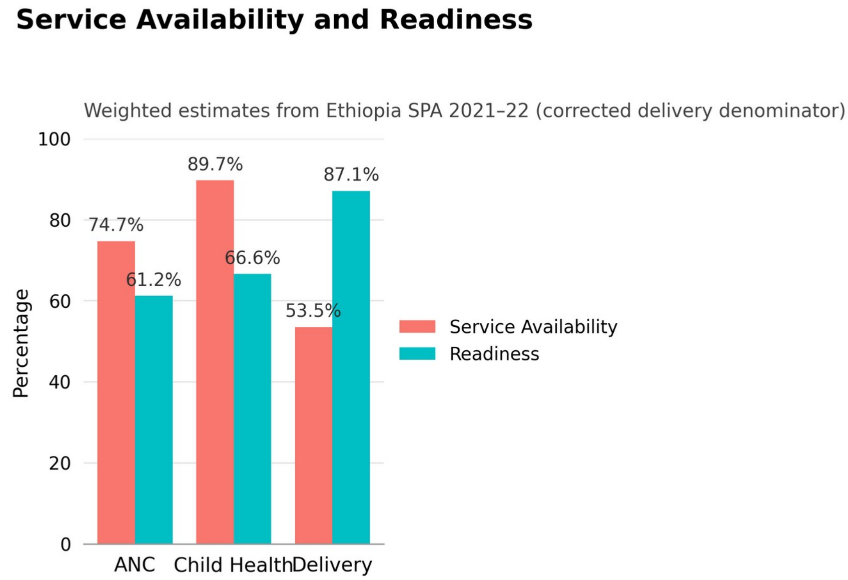
7. National Performance Results

Table 1. National Service Availability and Readiness, by Domain

Domain	Facilities with Service	Service Availability	Readiness	Readiness Gap
ANC	905	74.7%	61.2%	38.8 pp
Normal Delivery	650	53.5%*	87.1%	12.9 pp
Child Health	1,078	89.7%	66.6%	33.4 pp

* *Normal delivery availability is calculated among eligible facilities excluding health posts, consistent with the official SPA denominator. The figure of 650 represents facilities offering normal delivery and is not the denominator used to calculate the 53.5% availability estimate.*

Figure 1. Service Availability and Readiness, by Domain



Note on this figure: *the delivery bar in this chart reflects the validated availability estimate of 53.5% (facilities excluding health posts). An earlier all-facility calculation of 18.7% is not used anywhere in this report, consistent with the official SPA denominator described in Section 5.1.*

> **Interpretation:** Figure 1 shows an important disconnect between availability and readiness that runs in different directions across domains. Child-health services are available in 89.7% of facilities, but readiness is 23 percentage points lower (66.6%). ANC shows the same pattern even more sharply, with a 38.8-point gap between availability (74.7%) and readiness (61.2%) — the largest readiness gap of the three domains. Normal delivery inverts this pattern: availability is comparatively low (53.5%) but readiness among facilities that do offer delivery is the highest observed (87.1%), producing the smallest readiness gap (12.9 pp).

A service can be widely available and still poorly prepared to deliver it consistently. Availability alone, without readiness, overstates what a health system can actually do.

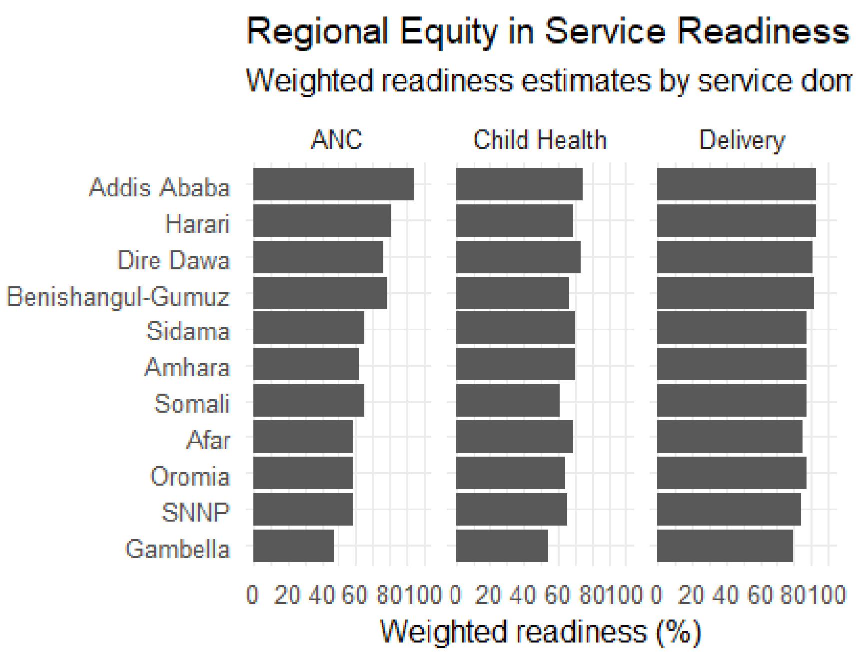
8. Equity Analysis

Equity was assessed across region, urban/rural residence, facility group, and managing authority. Equity gaps represent the difference between the highest and lowest weighted readiness estimate within each dimension.

Table 2. Equity Gaps in Readiness, by Domain and Dimension

Domain	Regional Gap	Urban–Rural Gap	Facility-Group Gap	Authority Gap
ANC	47.8 pp	18.9 pp	30.5 pp	28.7 pp
Normal Delivery	13.6 pp	4.6 pp	10.3 pp	3.0 pp
Child Health	20.8 pp	0.5 pp	22.0 pp	10.6 pp

Figure 2. Regional Equity in Service Readiness



Weighted readiness estimates by service domain (ANC, Child Health, Delivery), by region

> **Interpretation:** Figure 2 shows that regional variation in readiness is far from uniform across domains. ANC readiness spans nearly 48 percentage points between the best- and worst-performing regions, while delivery readiness across regions stays within a much narrower band. This means a single national readiness figure can conceal very different regional realities depending on which service is being monitored.

8.1 ANC — the largest equity problem

ANC shows the widest disparities of the three domains across every dimension examined: a 47.8-point regional gap, an 18.9-point urban–rural gap, a 30.5-point facility-group gap, and a 28.7-point managing-authority gap. This combination of low national readiness and the largest equity gaps makes ANC the most important equity priority identified in this analysis.

Table 3. ANC Weighted Readiness by Region

Region	ANC Readiness (%)
Addis Ababa	94.9

Region	ANC Readiness (%)
Harari	81.3
Benishangul-Gumuz	79.3
Dire Dawa	76.9
Sidama	65.7
Somali	65.2
Amhara	62.4
SNNP	58.9
Oromia	58.7
Afar	58.4
Gambella	47.1

Readiness ranges from 94.9% in Addis Ababa to 47.1% in Gambella — a 47.8-percentage-point regional gap.

8.2 Normal Delivery — comparatively equitable, but interpret cautiously

Delivery readiness shows the smallest gaps across all four dimensions (regional: 13.6 pp; urban–rural: 4.6 pp; facility group: 10.3 pp; authority: 3.0 pp), ranging regionally from 93.5% (Harari and Addis Ababa) to 79.9% (Gambella).

> **Caution:** Delivery subgroup sample sizes are very small for Health Posts (n = 5) and Clinics (n = 17). Facility-group comparisons involving these two groups should be interpreted with caution rather than treated as stable estimates, and the overall delivery equity pattern should be read with this limitation in mind.

8.3 Child Health — disparities driven by facility type and region, not residence

Child-health readiness varies substantially by region (20.8-pp gap, from 74.9% in Addis Ababa to 54.2% in Gambella) and by facility group (22.0-pp gap), but shows almost no difference between urban and rural facilities (0.5-pp gap). This indicates that where a facility is located matters less for child-health readiness than what type of facility it is and which region it operates in.

8.4 Facility Group and Managing Authority

Table 4. Readiness by Facility Group (%)

Facility Group	ANC	Normal Delivery	Child Health
Hospital	84.6	94.8	73.4
Health Center	68.1	86.2	79.0
Health Post	55.9	85.2 (n=5)	65.4
Clinic	86.4	84.5 (n=17)	57.1

Table 5. Readiness by Managing Authority (%)

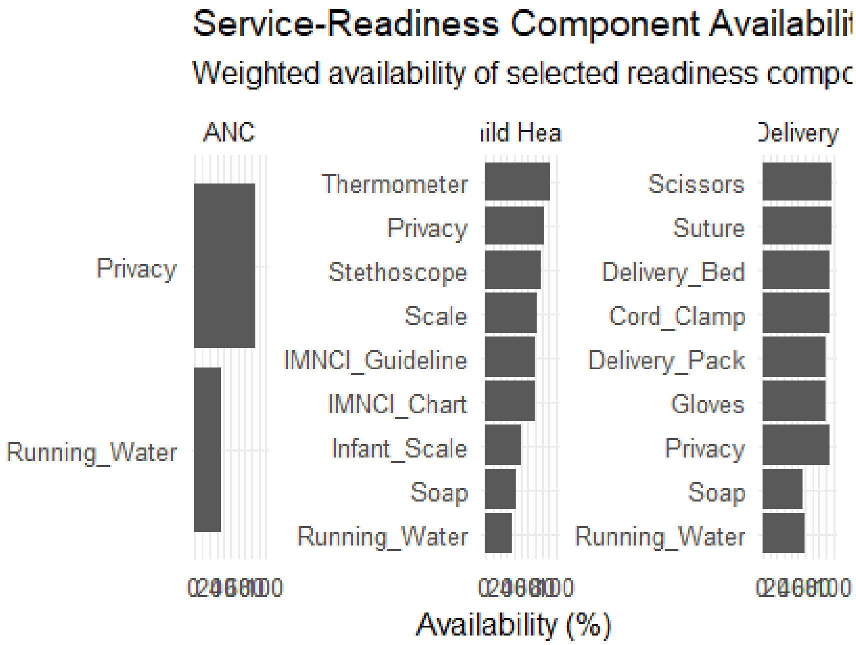
Authority	ANC	Normal Delivery	Child Health
Public	59.1	86.8	68.4
Non-government	87.8	89.8	57.7

These differences are descriptive. They should not be interpreted as evidence that facility type or managing authority causes differences in readiness; they may reflect a range of underlying structural and resourcing factors not examined in this analysis.

9. Readiness Bottleneck Analysis

Beyond overall readiness scores, examining individual components identifies exactly which inputs are missing most often — the specific operational bottlenecks behind each domain's readiness gap.

Figure 3. Service-Readiness Component Availability



Weighted availability of selected readiness components, by service domain (ANC, Child Health, Delivery)

> **Interpretation:** Figure 3 shows that the components with the lowest availability are consistently WASH-related — running water above all — rather than clinical equipment. Even in normal delivery, the domain with the highest overall readiness, running water (60.6%) and soap (58.7%) remain the two weakest components, trailing far behind items like scissors (98.3%) and delivery beds (96.6%).

9.1 ANC Components

Component	Weighted Availability
Privacy	84.6%
Running water	37.8%

With only two components in this constructed index, running water is the clear and dominant bottleneck for ANC readiness.

9.2 Normal Delivery Components

Component	Weighted Availability
Scissors	98.3%
Suture	96.9%
Cord clamp	96.6%
Delivery bed	96.6%
Privacy	95.1%

Component	Weighted Availability
Delivery pack	91.2%
Gloves	89.5%
Running water	60.6%
Soap	58.7%

The main bottleneck for normal delivery is soap (58.7%), with running water as an important secondary bottleneck (60.6%). Every other component exceeds 89% availability.

9.3 Child Health Components

Component	Weighted Availability
Thermometer	91.6%
Privacy	84.8%
Stethoscope	79.5%
Scale	73.5%
IMNCI guideline	70.9%
IMNCI chart	69.3%
Infant scale	50.7%
Soap	42.6%
Running water	38.7%

Running water (38.7%) is the main bottleneck for child health, with soap (42.6%) and infant scale availability (50.7%) as other major gaps.

9.4 Cross-Cutting WASH Bottleneck

Table 6. WASH Component Availability Across Domains

Component	ANC	Normal Delivery	Child Health
Running water	37.8%	60.6%	38.7%
Soap	—	58.7%	42.6%

> **Cross-cutting finding:** Running water and soap — the two most basic infection-prevention inputs — are the weakest readiness components in nearly every domain where they are assessed. These findings identify infrastructure and supply gaps that may undermine service readiness; they do not establish that WASH gaps cause specific outcomes, but they represent a clear, actionable priority for quality-improvement and M&E follow-up.

10. M&E Interpretation

Translating these findings into M&E language means thinking across four connected levels of the health system.

Input / Infrastructure Level

Running water, soap, and specific equipment items (infant scales, thermometers, delivery supplies) are the foundation on which service readiness is built. This analysis shows that infrastructure and supply gaps — not the absence of the service itself — are the binding constraint in most domains.

Service Readiness Level

ANC, delivery, and child-health readiness each tell a different story: ANC readiness is low and highly unequal; delivery readiness is comparatively strong among facilities offering the service; child-health readiness is moderate but unevenly distributed across facility types. Readiness needs to be tracked as its own indicator, separate from availability.

Equity Level

Regional differences, facility-type differences, and (to a lesser extent) urban/rural differences all shape whether a facility can deliver a ready service. ANC shows the starkest equity problem across every dimension examined; child-health disparities are concentrated in facility type and region rather than residence.

Program Monitoring Level

Routine M&E systems should monitor availability and readiness as complementary, not interchangeable, indicators, disaggregated by region and facility type wherever possible.

> **Indicators calculated in this project:** percentage of facilities offering ANC; percentage of eligible facilities offering normal delivery; percentage of facilities offering child-health services; mean constructed service-readiness score per domain; percentage of facilities with running water; percentage with soap; percentage with selected essential equipment; regional readiness gap; facility-type readiness gap.

> **Recommended for future routine monitoring (not calculated here):** these same indicators tracked over time at sub-national level, integrated into routine health-management information system (HMIS) dashboards — this project demonstrates the analytical approach rather than an established routine indicator set.

11. Priority Findings / Management Dashboard

A concise, management-oriented summary of the four priorities emerging from this analysis:

Priority 1 — ANC Readiness and Equity

High availability (74.7%) does not translate into adequate readiness (61.2%), and ANC shows the largest equity gaps of any domain across region, residence, facility group, and managing authority.

Priority 2 — WASH Infrastructure

Running water and soap are the most consistent readiness bottlenecks across ANC, delivery, and child health — a cross-cutting, addressable infrastructure gap.

Priority 3 — Child-Health Readiness

Near-universal service availability (89.7%) masks a substantial readiness gap (33.4 pp) and meaningful facility-type and regional disparities.

Priority 4 — Regional and Facility-Level Disparities

Readiness gaps are concentrated differently across domains — ANC by region and institution, child health by facility type and region — which argues for targeted rather than one-size-fits-all interventions.

12. Recommendations

The following programmatic priorities are suggested by these findings. They are not claims that any specific intervention will improve outcomes — they are priorities for further programmatic and quality-improvement attention.

1. Strengthen WASH infrastructure — running water and soap — as a cross-cutting priority across ANC, delivery, and child-health service points.

2. Improve availability of essential supplies, particularly for ANC (a two-component index already flags running water as critical) and child health (infant scales, soap).

3. Prioritize low-readiness regions — Gambella in particular shows the lowest readiness across all three domains.

4. Strengthen facility-level readiness monitoring, not just service-availability reporting, within routine HMIS systems.

5. Design facility-type-specific interventions, given that facility-group gaps are often as large as regional gaps (e.g., 30.5 pp for ANC, 22.0 pp for child health).

6. Integrate constructed readiness indicators into routine M&E dashboards alongside existing availability indicators.

7. Conduct targeted quality-improvement cycles in the lowest-readiness regions and facility types identified here.

8. Repeat facility readiness assessments periodically to track whether these gaps narrow over time.

13. Limitations

1. Cross-sectional design: this analysis captures a single point in time and cannot establish causality or track change over time.

2. Constructed readiness indices: the readiness scores used here are portfolio-specific constructions based on a defined set of tracer components and should not be confused with official SPA readiness indices.

3. ANC index limitation: the ANC readiness index includes only two components (privacy and running water), making it a comparatively simple measure relative to the nine-component delivery and child-health indices.

4. Delivery denominator: availability is calculated using the official eligible-facility denominator that excludes health posts, while readiness is assessed only among facilities that reported offering delivery — these are related but distinct populations.

5. Small subgroups: the delivery Health Post (n = 5) and Clinic (n = 17) subgroups are very small; facility-group comparisons for delivery should be interpreted cautiously.

6. Facility-level assessment: facility readiness reflects what a facility has available, not the quality of care an individual patient actually experiences.

7. Survey-weighted descriptive analysis: this project reports weighted descriptive and equity comparisons; no regression modeling, hypothesis testing, or causal inference was performed.

8. Missing/zero-weight facilities: 249 selected facilities did not contribute usable weighted analytical data and are not represented in any estimate in this report.

14. Conclusion

Ethiopia's health facilities show substantial service availability, but availability does not consistently translate into readiness. ANC presents the strongest combination of readiness and equity concerns, while child-health services show high availability but moderate readiness. Normal delivery services show lower availability but comparatively strong readiness among eligible facilities. Cross-cutting WASH gaps — particularly running water and soap — represent important operational bottlenecks.

> **M&E takeaway:** Routine monitoring should move beyond counting whether services exist and systematically track whether facilities are equipped and ready to deliver them, and whether readiness gaps are equitably distributed.

15. Technical Skills Demonstrated

Category	Skills
Programming & Data Management	R, RStudio, tidyverse, data cleaning, variable recoding
Survey Methods	Survey-weighted analysis, stratified sampling design, normalized survey weights
Indicator Development	M&E indicator construction, readiness index design, completeness rules
Analysis	Health-facility data analysis, equity analysis, component-level bottleneck analysis
Visualization	ggplot2, data storytelling, figure design for a non-technical audience
Domain Expertise	Public-health data interpretation, health-systems performance analysis
Communication	Evidence-based recommendations, M&E reporting for program and policy audiences

16. Reproducibility / Tools

The analysis workflow was developed entirely in R, using RStudio as the development environment. Key packages: tidyverse (data management), haven (reading SPSS/Stata-format survey data), labelled (managing variable and value labels), and ggplot2 (visualization). All estimates are survey-weighted using the normalized analysis weight described in Section 4.

Full project repository: *see Appendix B / GitHub README below.*

Appendix A. LinkedIn Project Description

What does it actually mean for a health facility to be “ready” to deliver care — not just to offer it? Using Ethiopia's Service Provision Assessment (SPA) 2021–22, a nationally representative, weighted survey of 1,158 health facilities, I analyzed antenatal care, normal delivery, and child-health services across three dimensions: service availability, service readiness, and equity.

The core finding: availability and readiness tell different stories depending on the service. ANC is widely available (74.7%) but readiness lags well behind (61.2%), with a striking 47.8-point gap in readiness between the best- and worst-performing regions. Child-health services are nearly universal (89.7% availability) but readiness sits at 66.6%, with facility-type disparities as large as regional ones. Normal delivery is less commonly available (53.5%, using the correct eligible-facility denominator) but shows the strongest readiness of the three (87.1%) once a facility offers it.

A cross-cutting pattern emerged at the component level: running water and soap — not advanced equipment — are the most consistent readiness bottlenecks across all three service domains.

This project applies survey-weighted analysis, constructed M&E indicators, equity gap analysis, and component-level bottleneck identification in R, translating a large facility dataset into findings that speak directly to program and policy decision-makers. It reflects the kind of analysis I want to keep doing: rigorous, transparent about its limitations, and built to inform action rather than just describe data.

Appendix B. GitHub README

The content below is formatted for direct use as this project's README.md on GitHub.

Health Facility Performance, Service Readiness, and Equity in Maternal and Child Health Service Delivery in Ethiopia

Overview

An M&E analysis of the Ethiopia Service Provision Assessment (SPA) 2021–22, examining service availability, service readiness, and equity for antenatal care (ANC), normal delivery, and child-health services across 1,158 survey-weighted health facilities.

Objectives

- Quantify national service availability and readiness for ANC, normal delivery, and child health.
- Identify equity gaps in readiness across region, residence, facility group, and managing authority.
- Identify component-level readiness bottlenecks.
- Translate findings into M&E indicators and programmatic recommendations.

Data Source

Ethiopia Service Provision Assessment (SPA) 2021–22. 1,407 facilities selected; 1,158 with positive analytical survey weights used in this analysis. Stratified by region and facility type; analysis weight = v005 ÷ 1,000,000 (normalized relative weight, not a population expansion factor).

Key Questions

1. How widely are ANC, normal delivery, and child-health services available?

2. How ready are facilities to provide these services?

3. How does readiness vary across region, residence, facility group, and authority?

4. Which components represent the largest readiness bottlenecks?

5. What are the implications for health-system monitoring and quality improvement?

Methods

- Survey-weighted proportions for service availability.
- Survey-weighted means for constructed readiness indices (ANC: 2 components; delivery and child health: 9 components each).
- Equity gaps: maximum minus minimum weighted readiness within each stratification dimension.
- Component-level bottleneck analysis of individual tracer items.
- Tools: R, RStudio, tidyverse, haven, labelled, ggplot2.

Key Findings

- ANC: 74.7% availability, 61.2% readiness, 38.8-pp readiness gap; largest equity gaps of the three domains (47.8 pp regional).
- Normal delivery: 53.5% availability (eligible facilities excluding health posts), 87.1% readiness, 12.9-pp readiness gap; most equitable domain, but delivery Health Post (n=5) and Clinic (n=17) subgroups are very small.
- Child health: 89.7% availability, 66.6% readiness, 33.4-pp readiness gap; facility-type and regional gaps far exceed the urban–rural gap (0.5 pp).

- Cross-cutting bottleneck: running water and soap are the weakest readiness components across nearly all domains.

Figures

- Figure 1 — Service Availability and Readiness by domain.
- Figure 2 — Regional Equity in Service Readiness.
- Figure 3 — Service-Readiness Component Availability.

M&E Implications

Availability and readiness should be tracked as complementary indicators, disaggregated by region and facility type. WASH infrastructure (running water, soap) is a cross-cutting priority for quality-improvement follow-up.

Limitations

- Cross-sectional; no causal inference.
- Constructed, not official SPA, readiness indices.
- ANC readiness index limited to 2 components.
- Delivery availability and readiness use different (related) denominators.
- Very small delivery Health Post / Clinic subgroups.
- Facility-level readiness ≠ patient-level quality of care.
- 249 facilities excluded for zero/missing weights.

Tools

R • RStudio • tidyverse • haven • labelled • ggplot2

Project Structure

data/ (not published — restricted SPA microdata) → scripts/ (cleaning, indicator construction, analysis, visualization) → figures/ (Figures 1–3) → report/ (this document)

Reproducibility

The SPA microdata are not redistributed in this repository due to data-use restrictions; scripts are provided to reproduce the analysis given access to the official Ethiopia SPA 2021–22 dataset.

Author

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Appendix C. LinkedIn Post

Does a health facility offering a service mean it's actually ready to deliver it? Usually not — and the gap between the two is bigger than I expected.

I analyzed Ethiopia's Service Provision Assessment (SPA) 2021–22, a nationally representative, weighted survey of 1,158 health facilities, to look at antenatal care (ANC), normal delivery, and child-health services from three angles: are they available, are facilities ready to deliver them, and is that readiness distributed fairly.

A few findings stood out:

- ANC is available in 74.7% of facilities, but readiness is only 61.2% — and readiness swings from 94.9% in Addis Ababa to 47.1% in Gambella.
- Child-health services are nearly universal (89.7% availability), but readiness sits at 66.6%, with facility-type gaps as large as regional ones.
- Normal delivery is less often available (53.5%, using the correct eligible-facility denominator) but shows the strongest readiness of the three domains (87.1%) once a facility offers it.
- Across every domain, the most consistent bottleneck wasn't advanced equipment — it was running water and soap.

The M&E lesson: counting whether a service exists tells you very little about whether a patient walking in today will be treated by a facility that's actually equipped for it. Availability and readiness need to be tracked side by side, and disaggregated by region and facility type, or real gaps stay hidden behind reassuring national averages.

All estimates are survey-weighted to reflect Ethiopia's stratified facility sampling design, and the full analysis — including equity gaps and component-level bottleneck analysis — was built in R.

I'd be glad to hear how others working in health-system M&E have approached the availability-versus-readiness distinction in their own contexts.